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## **COMPUTER ARCHITECTURE WK1**

### **1. Scenario Analysis:**

Memory access is an important process in computer architecture because the CPU needs to retrieve data and instructions from memory while running programs. For example, when a user opens a C++ calculator program, the operating system loads the program into RAM. The CPU then accesses the required instructions and data from RAM to execute the program. The CPU first checks its cache memory because it is faster than RAM. If the required information is not available in the cache, the CPU retrieves it from RAM. The data is transferred through the system bus between the CPU and memory. After processing, results may be stored back in RAM for later use. Efficient memory access improves computer performance because the CPU can process instructions faster. This scenario demonstrates the importance of different memory types, including cache, RAM, and storage, and shows how they work together to ensure that programs run smoothly and efficiently.

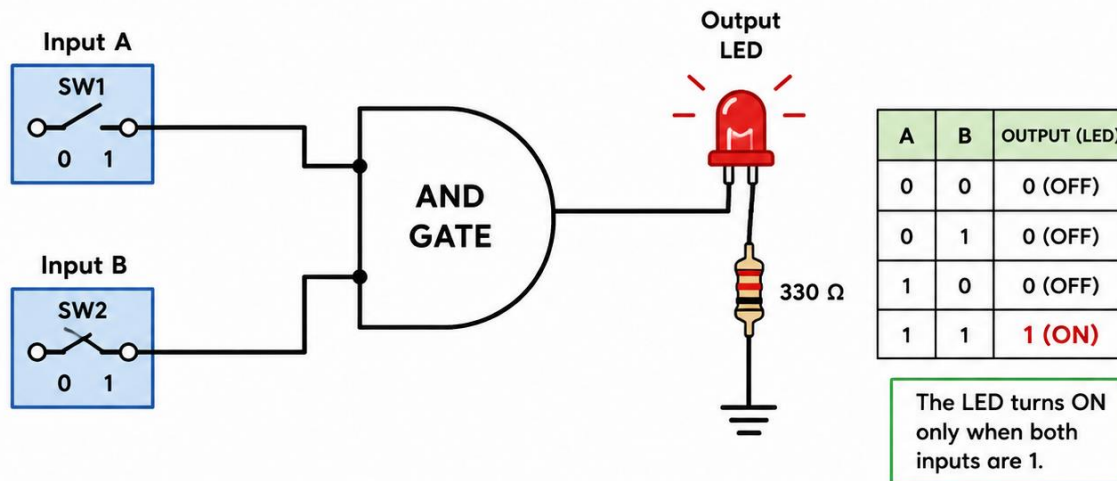
### **2.concept research**

The fetch-decode-execute cycle is a fundamental process used by the CPU to run computer programs. First, during the **fetch** stage, the CPU retrieves an instruction from memory using the program counter. Next, during the **decode** stage, the control unit interprets the instruction and determines what operation needs to be performed. Finally, during the **execute** stage, the CPU performs the required operation, such as calculating a value or moving data. The cycle then repeats continuously while a program is running. This process allows the CPU to execute instructions in the correct order and ensures that software functions properly and efficiently.

### **3.tool practice**

# SIMPLE DIGITAL CIRCUIT

## AND Gate Circuit



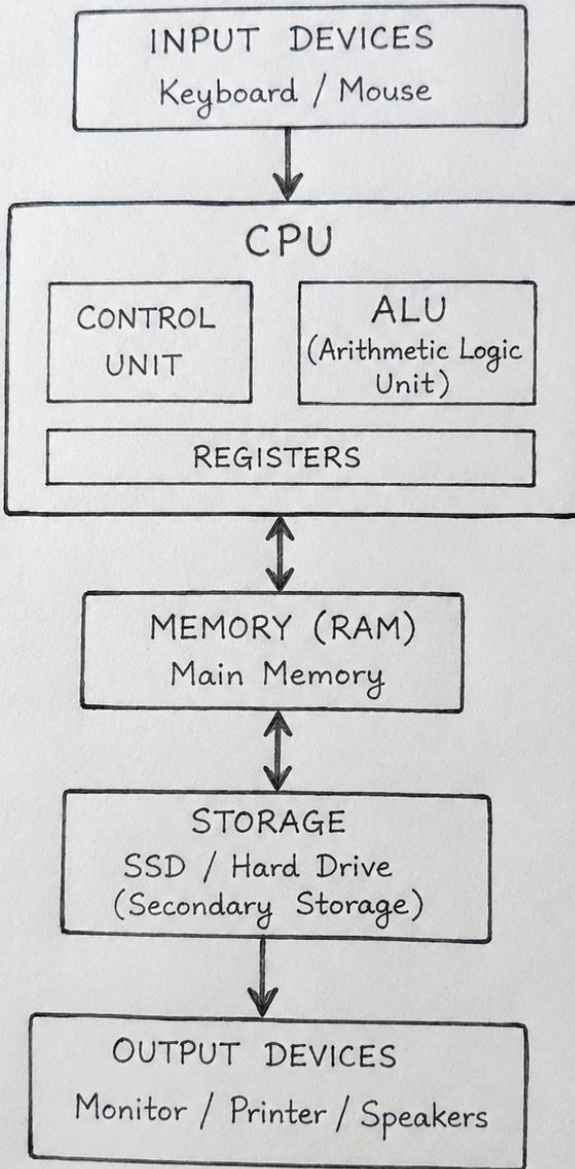
### REFLECTION (100 WORDS)

Creating a simple circuit using an AND gate helped me understand the basic principles of digital logic. I used two input switches, A and B, connected to an AND gate and an LED as the output. I tested different combinations of inputs to observe how the output changed. I learned that an AND gate produces an output of 1 only when both inputs are 1. Using Logisim/Canva made it easier to visualize how the components connect and communicate. This practical activity improved my understanding of logic gates, binary values, and digital circuits. It also showed me how simple circuits are used in computer systems.

## 4.architecture diagram

A computer architecture diagram shows how the main components of a computer communicate and work together. Input devices such as a keyboard and mouse allow users to enter data. The CPU is the main processing component and contains the Control Unit, Arithmetic Logic Unit (ALU), and registers. The Control Unit manages instructions, while the ALU performs calculations and logical operations. RAM temporarily stores data and instructions that the CPU needs while working. Storage devices such as SSDs and hard drives permanently store files and programs. Finally, output devices such as monitors and printers present the processed information to the user. Together, these components allow a computer to function efficiently.

## COMPUTER ARCHITECTURE DIAGRAM



## FLOW OF DATA

↓ Data flows from input to output

↔ Two-way communication